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WORKSHEET-MCQ

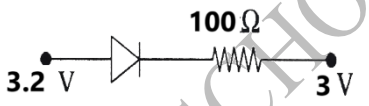
CLASS XII

PHYSICS

SEMICONDUCTOR ELECTRONICS

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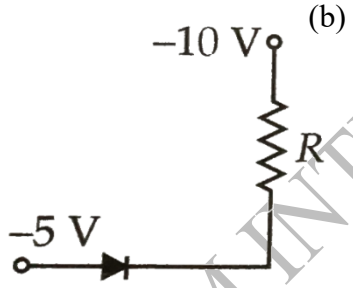
1. The current in the circuit shown in the figure considering an ideal diode is



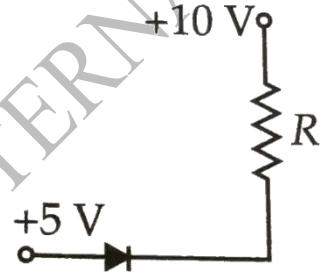
(a) A (b) 2×10^{-3} A (c) 200 A (d) 2×10^{-4} A

2. When a junction diode is reverse biased, the flow of current across the junction is mainly due to
   (a) diffusion of charges (b) the nature of material
   (c) drift of charges (d) both drift and diffusion of charges
3. A donor impurity results in
   (a) production of n- semiconductor (b) production of p- semiconductor
   (c) increase of resistance of the semiconductor (d) energy bands just above the filled valency
4. Which of the following p-n junction is forward biased?

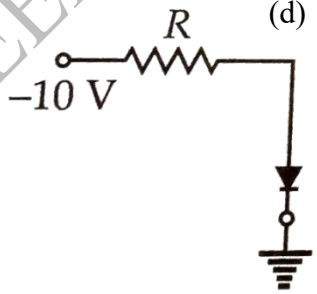
(a)



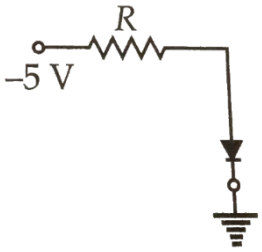
(b)



(c)



(d)

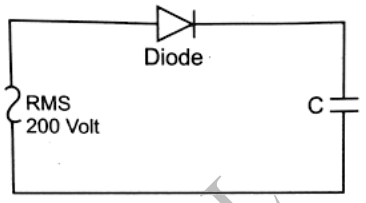


5. If no external voltage is applied across a p-n junction, there would be
   (a) no electric field across the junction
   (b) an electric field pointing from n- type to p- type side across the junction
   (c) an electric field pointing from p- type to n- type side across the junction

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(d) a temporary electric field during the formation of p-n junction that would subsequently disappear

6. In the circuit given in the figure, an a.c source of 200 V is connected through a diode D to a capacitor. The potential difference across the capacitor will be

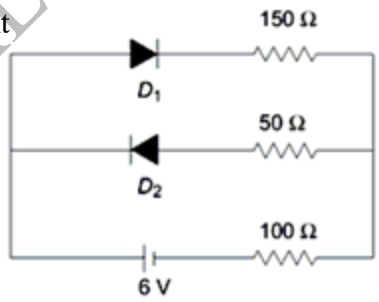


(a) 100 V (b) 283 V (c) 200 V (d) 310 V

7. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be

(a) 50 Hz (b) 70.7 Hz (c) 100 Hz (d) 25 Hz

8. The circuit shown in the figure contains two diodes each with a forward resistance of 50 Ω and with infinite backward resistance. If the battery voltage is 6 V, the current through the 100 Ω resistance is



(a) 0 A (b) 0.02 A (c) 0.03 A (d) 0.036 A

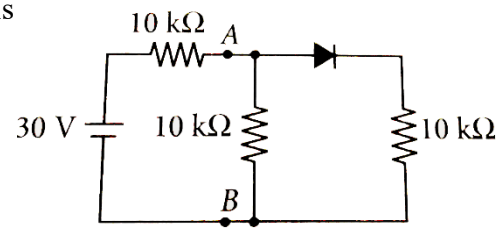
9. A Ge specimen is doped with Al. The concentration of acceptor atoms is $\approx 10^{21}$ atoms m^{-3}. Given that the intrinsic concentration of electron -hole pair is $\approx 10^{19}$ m^{-3}, the concentration of electrons in the specimen is

(a) 10^{17} m^{-3} (b) 10^{15} m^{-3} (c) 10^{4} m^{-3} (d) 10^{2} m^{-3}

10. The potential barrier in the depletion layer is due to

(a) ions (b) holes (c) electrons (d) forbidden band

11. In the given circuit, the potential difference between A and B is



(a) 0 V (b) 5 V (c) 10 V (d) 15 V

12. In semiconductors at room temperature

(a) the valence band is partially empty and the conduction band is partially filled
(b) the valence band is completely filled and the conduction band is partially filled
(c) the valence band is completely filled
(d) the conduction band is completely empty

13. Reverse bias applied to a junction diode

(a) lowers the potential barrier (b) raises the potential barrier

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(c) increases the majority carrier current (d) increases the minority carrier current

For questions 14 to 18, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

14. **Assertion:** A pure semiconductor has negative temperature coefficient of resistance.
Reason: On raising the temperature, more charge carriers are released, conductance increases and resistance decreases.
15. **Assertion:** An n-type semiconductor has a large number of electrons but still it is electrically neutral.
Reason: An n-type semiconductor is obtained by doping an intrinsic semiconductor with a pentavalent impurity.
16. **Assertion:** The drift current in a p-n junction is from the n-side to the p-side.
Reason: It is due to free electrons only.
17. **Assertion:** The dominant mechanism for motion of charge carriers in forward and reverse biased silicon p- n junction are drift in both forward and reverse bias.
Reason: In reverse biasing, no current flow through the junction.
18. **Assertion:** The number of electrons in a P-type silicon semiconductor is less than the number of electrons in a pure silicon semiconductor at room temperature.
Reason: It is due to law of mass action.

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